I. Cloud Computing Fundamentals

1. **What is Cloud Computing with example?**
    a. **Definition:** On-demand delivery of IT resources (servers, storage, databases) over the internet with pay-as-you-go pricing.
    b. **Example:** Using **Google Drive** or **OneDrive** to store files (SaaS) or renting a virtual server on **AWS EC2** (IaaS).
2. **Who is the father of cloud computing?**
    a. **J.C.R. Licklider** is considered the father of cloud computing, who in the 1960s proposed the idea of a global "intergalactic computer network".
3. **Basic characteristics of cloud computing?**
    a. **On-Demand Self-Service:** Resources available without human interaction with the provider.
    b. **Broad Network Access:** Accessible via internet on any device.
    c. **Resource Pooling:** Multi-tenant model where resources are shared among many users.
    d. **Rapid Elasticity:** Scalable to meet demand.
    e. **Measured Service:** Pay-per-use model.
4. **Vertical and Horizontal scaling?**
    a. **Vertical (Scaling Up):** Increasing the capacity of a single resource (e.g., adding more RAM to a virtual machine).
    b. **Horizontal (Scaling Out):** Adding more resources to the pool (e.g., adding more virtual machines to a cluster).
5. **Services provided by Cloud Computing?**
    a. **IaaS (Infrastructure as a Service):** Compute/Storage (e.g., AWS EC2).
    b. **PaaS (Platform as a Service):** Development environment (e.g., Google App Engine).
    c. **SaaS (Software as a Service):** Finished apps (e.g., Salesforce).
6. **Deployment models in Cloud Computing?**
    a. **Public Cloud:** Services available to the public over the internet (e.g., AWS).
    b. **Private Cloud:** Exclusive to one business.
    c. **Hybrid Cloud:** Combination of private and public.
    d. **Community Cloud:** Shared by organizations with common concerns.
7. **Platforms for Large Scale Cloud Computing?**
    a. AWS (Amazon Web Services), Microsoft Azure, Google Cloud Platform (GCP).
8. **Large cloud providers and databases?**
    a. **Providers:** Amazon, Microsoft, Google, IBM, Oracle.
    b. **Databases:** Amazon RDS/DynamoDB, Azure SQL, Google Cloud SQL, MongoDB Atlas.

9. **Difference between cloud and traditional data centers?**
    a. **Cloud:** Pay-as-you-go, high scalability, managed by provider, flexible pricing.
    b. **Traditional:** High upfront capital expense, limited scalability, managed internally.
10. **Components required in cloud computing?**
    a. Client Infrastructure, Application, Service, Runtime Cloud, Storage, Infrastructure, Management, Security.
11. **Platforms of cloud architecture?**
    a. Front-end (User side) and Back-end (Server side) connected via the internet.
12. **What is a cloud service?**
    a. Any service provided through the internet (SaaS, PaaS, IaaS) to simplify IT operations.
13. **Basic clouds?**
    a. Public, Private, Hybrid.
14. **Issues with Cloud Computing?**
    a. Security/Privacy, Technical Downtime, Vendor Lock-in, Internet Dependency.

II. Cloud Providers & Virtualization [1]

15. **Services provided by Windows Azure?**
    a. Virtual Machines, SQL Database, App Service, Active Directory, Azure Storage, Azure Functions.
16. **What is Cloud Computing Architecture?**
    a. Components (frontend/backend) required to manage, store, and distribute data.
17. **Services provided by AWS?**
    a. EC2 (Compute), S3 (Storage), RDS (Database), Lambda (Serverless).
18. **Services provided by Microsoft?**
    a. Azure SQL, Azure VM, Azure AI, Azure DevOps.
19. **What is Virtualization?**
    a. Technology to create virtual versions of physical resources (servers, storage) to abstract hardware.
20. **What is an EC2 instance?**
    a. A virtual server in AWS (Elastic Compute Cloud) that provides scalable computing capacity.
21. **What is AMI in EC2?**
    a. **Amazon Machine Image (AMI):** A template providing information (OS, application server) required to launch an instance.
22. **How to launch an on-demand EC2 instance?**
    a. Sign in to AWS Console -> EC2 Dashboard -> Launch Instance -> Select AMI -> Choose Instance Type -> Configure Security Group -> Launch. [1, 2, 3, 4, 5]

III. Google App Engine & Salesforce [1]

23. What is Google App Engine?

a. A PaaS platform to build and host web applications in Google-managed data centers.

24. Advantages of Google App Engine?

a. Auto-scaling, easy deployment, supports popular languages (Python, Java), automatic load balancing.

25. Steps to install and configure Google App Engine?

a. Install SDK -> Create app -> Write app.yaml -> Run locally -> Deploy via command line.

26. What is Apex?

a. A proprietary, strongly-typed, object-oriented programming language by Salesforce used to execute flow and transaction control statements on the Lightning Platform.

27. Features of Apex?

a. Data-focused, object-oriented, easy to test, integrates with database.

28. Applications of Apex?

a. Creating web services, Triggers (similar to DB triggers), custom controllers.

29. Apex Code Development Tools?

a. Developer Console, Force.com IDE (Eclipse), VS Code with Salesforce Extensions.

30. Steps to create application using Apex?

a. Define Data Model -> Write Apex Class/Trigger -> Test the Class -> Deploy to Salesforce.

31. What is Salesforce.com Inc.?

a. A cloud-based CRM (Customer Relationship Management) software provider.

32. What is Lightning Platform?

a. A PaaS for developing Salesforce applications.

33. How to create Custom Application using Salesforce Classic?

a. Setup -> App Setup -> Create -> Apps -> New.

34. Difference between custom and console application in Salesforce?

a. **Custom:** Standard navigation.
b. **Console:** Tab-based layout for multitasking.

35. Steps to create custom application in Salesforce?

a. Click Setup -> Build -> Apps -> New -> Enter details -> Assign profiles.

IV. Architecture, Trends & Applications

36. What is hypervisor in Cloud Computing?

a. Software that creates and runs virtual machines (VMs) by partitioning a physical server.

37. Explain Load Balancing?

a. Distributing network traffic across multiple servers to ensure no single server bears too much load, enhancing reliability.

38. Open-source cloud computing platforms/databases?

a. OpenStack, Apache CloudStack, Eucalyptus, MongoDB (NoSQL), MySQL.

39. Recent trends of Cloud Computing?

a. Serverless Computing, Multi-cloud strategy, Edge computing, AI/ML integration.

40. Applications of Cloud Computing?

a. Data storage (Drive), Healthcare analytics, E-commerce, Financial services, Gaming.

AI-Oral Question

1. Differentiate between DFS and BFS Algorithms.

DFS (Depth First Search)	BFS (Breadth First Search)
Explores depth first	Explores level by level
Uses Stack	Uses Queue
May not give shortest path	Gives shortest path in unweighted graph
Requires less memory	Requires more memory
Good for deep solutions	Good for nearest solutions

2. Write down the Time and Space Complexity of DFS and BFS.

Algorithm	Time Complexity	Space Complexity
DFS	O(V + E)	O(V)
BFS	O(V + E)	O(V)

Where:

- **V** = Number of vertices
- **E** = Number of edges

3. Which data structure is used by DFS and BFS?

- **DFS** uses **Stack** (or recursion stack).
- **BFS** uses **Queue**.

4. Differentiate between Best First Search and A* Algorithm.

Best First Search	A* Algorithm
Uses heuristic function h(n)	Uses f(n) = g(n) + h(n)
Chooses node closest to goal	Chooses optimal path
Faster but may not be optimal	Optimal and complete
Ignores path cost	Considers path cost

Where:

- **g(n)** = actual cost
- **h(n)** = heuristic cost

5. Solve this problem using A* Algorithm.

Steps of A* Algorithm:

1. Start from initial node.
2. Calculate:

*f(n)=g(n)+h(n)f(n)=g(n)+h(n)*f(n)=g(n)+h(n)

3. Choose node with minimum f(n).
4. Expand node.
5. Repeat until goal state is reached.

Example:

If:

- g(n) = 4

- h(n) = 3

Then:

*f(n)=4+3=7f(n)=4+3=7*f(n)=4+3=7

The node with smallest f(n) is selected first.

6. What is the drawback to solve 8 Puzzle problem with a non-heuristic method?

Drawbacks:

- Very high time complexity.
- Explores unnecessary states.
- Consumes more memory.
- Solution becomes very slow.

Heuristic methods reduce search space and improve efficiency.

7. What is the time and space complexity of selection sort?

Case	Time Complexity
Best	$O(n^2)$
Average	$O(n^2)$
Worst	$O(n^2)$

Space Complexity:

- **O(1)** (in-place sorting)

8. If an array is [6, 1, 9, 10], sort the list by selection sort step wise.

Initial Array:

[6, 1, 9, 10]

Pass 1:

- Smallest element = 1
- Swap 6 and 1

[1, 6, 9, 10]

Pass 2:

- Smallest in remaining = 6
- No swap

[1, 6, 9, 10]

Pass 3:

- Smallest in remaining = 9
- No swap

Final Sorted Array:

[1, 6, 9, 10]

9. What is the maximum number of comparisons in one iteration for an array of size N?

Maximum comparisons in first iteration:

*N−1N-1*N−1

For total comparisons in selection sort:

*N(N−1)2\frac{N(N-1)}{2}*2N(N−1)

10. Draw comparison chart with other sorting techniques with respect to time and space complexity.

Sorting Algorithm	Best	Average	Worst	Space
Bubble Sort	O(n)	$O(n^2)$	$O(n^2)$	O(1)
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	O(1)
Insertion Sort	O(n)	$O(n^2)$	$O(n^2)$	O(1)
Merge Sort	O(n log n)	O(n log n)	O(n log n)	O(n)
Quick Sort	O(n log n)	O(n log n)	$O(n^2)$	O(log n)

11. Explain Complexity.

Complexity measures efficiency of an algorithm.

Types:

1. **Time Complexity** → Time taken by algorithm.
2. **Space Complexity** → Memory used by algorithm.

Example:

- O(1) → Constant
- O(n) → Linear
- $O(n^2)$ → Quadratic

12. What is a Stable Sort Algorithm? Whether selection sort is stable?

Stable Sort:

A sorting algorithm is stable if equal elements maintain their original order.

Example:

5A 3 5B

After stable sort:

3 5A 5B

Is Selection Sort Stable?

- **No**, Selection Sort is not stable.

13. Which are the constraints required to solve the N Queen Problem?

Constraints:

1. Only one queen per row.
2. Only one queen per column.
3. No two queens on same diagonal.

14. Compare Backtracking and Branch and Bound methods.

Backtracking	Branch and Bound
Used for decision problems	Used for optimization problems
Uses DFS approach	Uses BFS/Best-first approach
Prunes invalid solutions	Prunes non-optimal solutions
Finds feasible solution	Finds best solution

15. What do you mean by Constraint Satisfaction Problem?

A problem where solution must satisfy given constraints.

Components:

1. Variables
2. Domains
3. Constraints

Examples:

- Sudoku
- N Queen Problem

- Map Coloring

16. What is the use of a chatbot?

Uses of chatbot:

- Customer support
- Answering queries
- Booking services
- Education assistance
- Banking support
- E-commerce support

17. Explain dialog flow in detail.

Dialog flow is conversation structure between user and chatbot.

Components:

1. **Intent** → User purpose
2. **Entity** → Important data
3. **Context** → Conversation state
4. **Response** → Bot reply

Flow:

User Input → Intent Detection → Processing → Response Generation

18. What are the requirements for developing a chatbot?

Requirements:

- NLP (Natural Language Processing)
- Training data
- Backend server
- Database
- APIs
- Dialog management
- User Interface

19. How do you evaluate chatbot performance?

Methods:

- Accuracy
- Response time
- User satisfaction
- Intent recognition rate
- Error rate
- Conversation completion rate

20. How do I improve my chatbot accuracy?

Ways to improve:

- Use better training data
- Train with more examples
- Improve NLP model
- Add context handling
- Regular testing
- Continuous learning and feedback analysis